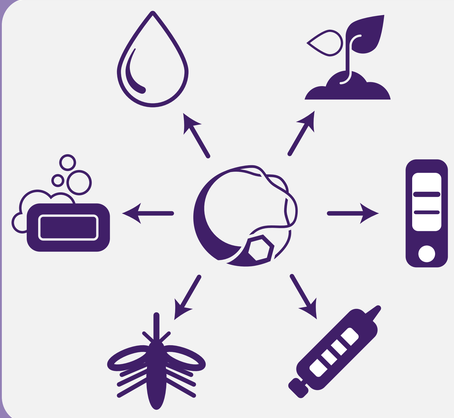
From Lab to Field: How Synthetic Biology Can Serve Global Development

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Overview



Five Deployment Readiness Factors

1. **Cold Chain** — Independence from refrigeration
2. **Partnership Model** — Local co-design vs. co-execution
3. **Regulatory Path** — Environmental & Political
4. **Cost Target** — <$5–10/use for most cases
5. **Usability Design** — Non-expert interpretable output

Motivation

Experience: Engineers Without Borders

Working on water infrastructure projects in Central America with EWB, the most important lesson wasn't technical: **sustainable development requires community independence.** Good solutions are ones communities can operate, maintain, and adapt themselves. A solution that requires outside experts to return is not a sustainable solution.

The Gap

Synthetic biology is producing tools directly relevant to the problems many communities face — water quality, food safety, disease. But the lab-to-field pathway is often realized through opportunistic partnerships and most scientific research is done within an academic bubble without local context.

Research Questions

1. What makes a synthetic biology technology field-deployable vs. lab-only?
2. What are the barriers to translating promising synbio research into real-world global development applications?
3. Where are the missing connections for technologies to reach communites that would benefit?

Methods: Multi-source research synthesis

Case studies · Literature review · Stakeholder perspectives · Deployment framework

- Case study analysis: 3 deployed or in-development synbio projects in global development contexts, analyzed using a consistent 5-factor deployment readiness framework
- Literature synthesis across synthetic biology, global health, humanitarian engineering, and development practice

Stakeholder interviews conducted across three groups:

Synthetic biology researchers:
Julius Lucks (Northwestern), Keith Pardee (University of Toronto), Tibebu Habtewold (ICL)

Global development practitioners:
Dickson Wilson Lwetoijera (Ifakara Health Institute, Tanzania), Sera Young (Northwestern)

Humanitarian engineering:
EWB Chicagoland chapter leadership, EWB-USA leadership

Interview themes:
- partnership structure
- translation barriers
- community ownership
- design for independence

New skills gained:
- Stakeholder interview methodology
- Interdisciplinary synthesis
- Technology translation framework development
- Science communication to non-research audiences

Synbio tools can address critical global development needs, but translation requires deliberate local partnership

Positive outlook: There are multiple cases that highlight how effective synthetic biology can be at changing peoples' lives

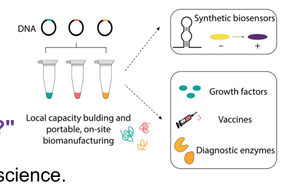
Field deployability and local engagement should be designed in from day one, not retrofitted

Critical partnership gaps exist between proven synbio platforms and development organizations that need them

For CSB Researchers:

Ask these questions throughout the scientific process:

1."Who is defining the problem?"
The most successful tools here share a common trait:
The in-country partnership was built alongside the tool
You cannot tell a community what their problem is



2. "Does this create independence or dependency?"
Transferring tools from academic labs is faster.
Building capacity produces communities that can own the science.

3. "How does the local context change my design constraints?"
Include social scientists in design
Explicitly consider how your design would change outside a well-equipped lab

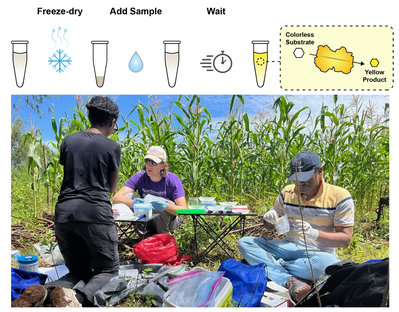
Where the connections aren't built yet:
- Agriculture: Cell-free biosensor tech proven for small molecules
  - ~25% of Sub-Saharan African cereal crops contaminated with aflatoxin annually with no farm-level test
- Sanitation: Fecal biosensors
  - EWB lacks community tools to verify latrine performance post-deployment

References:

Thavarajah et al., npj Clean Water (2023); Pardee et al. Cell (2016); Ribeiro et al medRxiv (2025); Karlikow et al., Nature Biomedical Engineering (2022); Habtewold et al., Nature (2025).

1 ROSALIND — Cell-Free Water Biosensing

Lucks Lab, Northwestern · Field study: Kenya · 2023



98.2% of participants correctly interpreted test

89.5% accuracy vs. gold-standard fluoride photometer, Kenya

57 water samples tested 36 households, Nakuru Co.

17 contaminants detectable by platform including lead (Chicago pilot ongoing)

Technology

- Cell-free RNA biosensing platform
- Freeze-dried — no cold chain required
- Fluorescent visual readout, interpretable
  - <$10/test

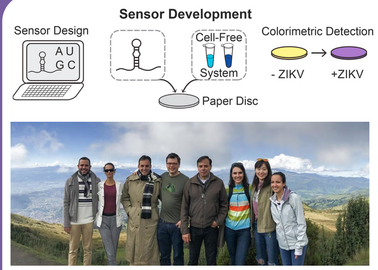
What Made It Field-Deployable

- No cold chain required
- Cell-free system: no GMO release
- Social science from project inception
- Usability was a core metric
- Kenyan co-investigators embedded

Technology was designed at Northwestern. A path to local production is a logical step toward full community independence

2 Paper Diagnostics for Viral Diseases

Pardee / Collins Labs · Clinical trial: Recife, Brazil



98.5% accuracy vs. CDC PCR gold standard

$5/test no PCR supply chain needed

Lab → Field → Startup En Carta Diagnostics (Paris)

268 patient samples tested during active outbreak

Technology

- Cell-free RNA sensors
  - freeze-dried onto paper,
  - colorimetric readout
- PLUM reader: ~$500, portable
- Programmable to pathogen sequences



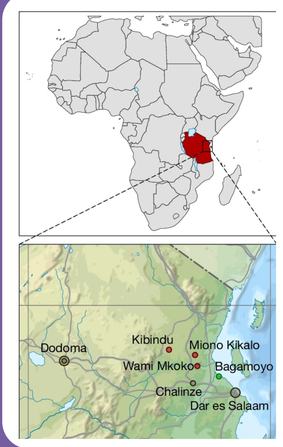
What Made It Field-Deployable

- Fiocruz as partner from planning stage
  - in-country expertise shaped design
- Tested during 2015–2016 outbreak
- No PCR supply chain dependency
- Visual readout, minimal training

Why this case matters:
Tested during active outbreak
Demonstrates real-world urgency tests
Complete arc: lab → field → commercial

3 Gene Drive Mosquitoes

Imperial College London + Tanzanian partners · 2025



1st gene drive-capable mosquito engineered in Africa, by African scientists

0% salivary gland infection in engineered mosquitoes

3 villages, Pwani year-round surveillance with locally infected children

~94% transgene inheritance rate with Cas9

Technology

- Engineered to express antimicrobial peptides
- Tested against locally circulating parasite from Tanzanian children
- not lab-cultured strains

What Made It Field-Ready

- Engineering done in Tanzania
- Insectary built at Ifakara Health Institute, Bagamoyo
- Local strains engineered against local *P. falciparum* isolates
- Tanzanian scientists are co-creators
- Community consent and public engagement conducted throughout

Nuance: Ecosystem intervention — modification designed to spread through wild populations. Highest regulatory and ethical stakes of any technology in this analysis. Field release has not yet occurred.

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